ElderCareAI Unit Testing

One hundred selected unit cases cover the essential caregiver, watch, health, care plan, alert, and administrator behavior. Results are from the Jest run on 9 November 2026.

Proponent	Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Result	Actual Result	Result
	Authentication Module	Google Sign-In	11/09/2026	CASE-001	Open the login screen.	Google sign-in button is visible.	Performed as expected	Passed
	Authentication Module	Google Sign-In	11/09/2026	CASE-002	Tap Google sign-in and return a successful mocked result.	Sign-in starts and the app redirects to the route resolver.	Performed as expected	Passed
	Authentication Module	Google Sign-In	11/09/2026	CASE-003	Return a failed mocked sign-in result.	Error appears and the login screen remains open.	Performed as expected	Passed
	Authentication Module	Session Routing	11/09/2026	CASE-004	Open login with an existing caregiver session.	App redirects to the introduction screen.	Performed as expected	Passed
	Onboarding Module	Setup Routing	11/09/2026	CASE-005	Open the app as a new caregiver.	Introduction screen opens first.	Performed as expected	Passed
	Onboarding Module	Setup Routing	11/09/2026	CASE-006	Leave caregiver information incomplete.	Caregiver setup remains the next step.	Performed as expected	Passed
	Onboarding Module	Setup Routing	11/09/2026	CASE-007	Leave older-adult information incomplete.	Older-adult setup remains the next step.	Performed as expected	Passed
	Onboarding Module	Setup Routing	11/09/2026	CASE-008	Leave wearable setup incomplete.	Wearable setup remains the next step.	Performed as expected	Passed
	Onboarding Module	Setup Routing	11/09/2026	CASE-009	Leave location consent incomplete.	Home access is deferred until location setup.	Performed as expected	Passed
	Onboarding Module	Setup Routing	11/09/2026	CASE-010	Complete every onboarding step.	Home screen becomes the next route.	Performed as expected	Passed
	Form Components Module	Form Field	11/09/2026	CASE-011	Render a field with validation feedback.	Error appears beside the related input.	Performed as expected	Passed
	Form Components Module	Form Field	11/09/2026	CASE-012	Render the authenticated email field.	Email value cannot be edited.	Performed as expected	Passed
	Form Components Module	Choice Field	11/09/2026	CASE-013	Select an available choice.	Component reports the selected option.	Performed as expected	Passed
	Caregiver Profile Module	Caregiver Setup	11/09/2026	CASE-014	Submit missing required caregiver details.	Field errors appear and no profile is saved.	Performed as expected	Passed
	Caregiver Profile Module	Caregiver Setup	11/09/2026	CASE-015	Enter an invalid caregiver phone number.	Profile save is blocked.	Performed as expected	Passed
	Caregiver Profile Module	Caregiver Setup	11/09/2026	CASE-016	Enter a caregiver age below the allowed minimum.	Form rejects the age and does not save.	Performed as expected	Passed
	Caregiver Profile Module	Caregiver Setup	11/09/2026	CASE-017	Submit valid caregiver details.	Profile saves and older-adult setup opens.	Performed as expected	Passed
	Caregiver Profile Module	Caregiver Setup	11/09/2026	CASE-018	Return a mocked database error during caregiver save.	Error appears and setup does not advance.	Performed as expected	Passed
	Older Adult Profile Module	Older Adult Setup	11/09/2026	CASE-019	Submit missing required older-adult details.	Required-field guidance appears.	Performed as expected	Passed
	Older Adult Profile Module	Older Adult Setup	11/09/2026	CASE-020	Enter implausible weight and height.	Profile save is blocked.	Performed as expected	Passed
	Older Adult Profile Module	Older Adult Setup	11/09/2026	CASE-021	Select a birth date from the calendar.	The selected date is saved and determines the older adult's age.	Performed as expected	Passed
	Older Adult Profile Module	Older Adult Setup	11/09/2026	CASE-022	Enter an invalid emergency phone number.	Profile save is blocked.	Performed as expected	Passed
	Older Adult Profile Module	Older Adult Setup	11/09/2026	CASE-023	Submit valid older-adult details.	Profile saves and wearable setup opens.	Performed as expected	Passed
	Older Adult Profile Module	Older Adult Setup	11/09/2026	CASE-024	Return a mocked database error during older-adult save.	Error appears and setup does not advance.	Performed as expected	Passed

Proponent	Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Result	Actual Result	Result
	Vitals Module	Vital Card	11/09/2026	CASE-025	Render a patient vital card.	Patient reading and timestamp are shown.	Performed as expected	Passed
	Vitals Module	Vital Card	11/09/2026	CASE-026	Tap the patient vital card.	Reading detail action is called.	Performed as expected	Passed
	Vitals Module	Vital Card	11/09/2026	CASE-027	Render a full-width vital card.	Card uses the readable row layout.	Performed as expected	Passed
	Health Insights Module	Insight Generation	11/09/2026	CASE-028	Calculate insight with no reading history.	Waiting insight is returned.	Performed as expected	Passed
	Health Insights Module	Insight Generation	11/09/2026	CASE-029	Analyze a low oxygen reading.	Critical oxygen insight is returned.	Performed as expected	Passed
	Health Insights Module	Insight Generation	11/09/2026	CASE-030	Analyze short sleep duration.	Sleep warning insight is returned.	Performed as expected	Passed
	Health Insights Module	Trend Analysis	11/09/2026	CASE-031	Request a trend with fewer than four distinct readings.	Prediction waits for more data.	Performed as expected	Passed
	Health Insights Module	Trend Analysis	11/09/2026	CASE-032	Analyze four distinct readings.	Labeled statistical trend is returned.	Performed as expected	Passed
	Health Insights Module	Anomaly Detection	11/09/2026	CASE-033	Analyze abnormal heart rate and oxygen together.	Both measurements are flagged.	Performed as expected	Passed
	Health Insights Module	Anomaly Detection	11/09/2026	CASE-034	Analyze a normal latest reading.	No anomaly is claimed.	Performed as expected	Passed
	Health Insights Module	HRV Insight	11/09/2026	CASE-035	Calculate an HRV insight.	Raw RMSSD is reported in milliseconds.	Performed as expected	Passed
	Elle Assistant Module	Chat Screen	11/09/2026	CASE-036	Open Elle chat.	Non-medical assistant disclaimer is visible.	Performed as expected	Passed
	Elle Assistant Module	Message Input	11/09/2026	CASE-037	Try to send a blank message.	Assistant request is not called.	Performed as expected	Passed
	Elle Assistant Module	Chat Reply	11/09/2026	CASE-038	Send one question with a mocked assistant reply.	Question and reply render once.	Performed as expected	Passed
	Elle Assistant Module	Chat Error	11/09/2026	CASE-039	Return a mocked service failure.	Friendly message appears without raw Bad Request text.	Performed as expected	Passed
	Elle Assistant Module	New Chat	11/09/2026	CASE-040	Start a new chat after previous messages.	Old replies clear and welcome text returns.	Performed as expected	Passed
	Care Plan Module	Medication List	11/09/2026	CASE-041	Open an empty medication list.	Clear empty state appears.	Performed as expected	Passed
	Care Plan Module	Medication Form	11/09/2026	CASE-042	Save a medication without a name.	Save is rejected with validation feedback.	Performed as expected	Passed
	Care Plan Module	Medication Form	11/09/2026	CASE-043	Choose a medication reminder time with the picker.	The selected time appears in the form without manual entry.	Performed as expected	Passed
	Care Plan Module	Caregiver Note	11/09/2026	CASE-044	Save an empty caregiver note.	Empty note is rejected.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-045	Check an as-needed medication.	No scheduled dose is shown.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-046	Check a medication taken today.	Dose is marked taken.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-047	Check yesterday's or another medication's log.	It is not counted as taken today.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-048	Check a medicine without reminder time.	Status is unscheduled.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-049	Check a dose due within 30 minutes.	Status is due now.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-050	Check a dose due within three hours.	Status is due soon.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-051	Check a later scheduled dose.	Status is upcoming.	Performed as expected	Passed
	Medication Tracking Module	Dose Status	11/09/2026	CASE-052	Check a dose whose time has passed.	Status says due earlier.	Performed as expected	Passed
	Medication Tracking Module	Dose Sorting	11/09/2026	CASE-053	Sort missed and upcoming doses.	Missed doses appear first.	Performed as expected	Passed

Proponent	Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Result	Actual Result	Result
	Reminders Module	Time Validation	11/09/2026	CASE-054	Enter valid 24-hour reminder times.	Valid times are accepted.	Performed as expected	Passed
	Reminders Module	Time Validation	11/09/2026	CASE-055	Enter malformed or out-of-range times.	Invalid times are rejected.	Performed as expected	Passed
	Reminders Module	Reminder Schedule	11/09/2026	CASE-056	Calculate the second daily reminder from 20:30.	Second reminder time wraps to 08:30.	Performed as expected	Passed
	Reminders Module	Medication Alert	11/09/2026	CASE-057	Schedule an as-needed medication.	No timed alert is created.	Performed as expected	Passed
	Reminders Module	Appointment Alert	11/09/2026	CASE-058	Deny notification permission.	App does not claim an appointment alert was set.	Performed as expected	Passed
	Reminders Module	Appointment Alert	11/09/2026	CASE-059	Permit an appointment alert for a patient.	Alert text names the patient.	Performed as expected	Passed
	Doctor Contact Module	Doctor Settings	11/09/2026	CASE-060	Open Settings without a saved doctor.	Empty doctor contact state appears.	Performed as expected	Passed
	Doctor Contact Module	Doctor Settings	11/09/2026	CASE-061	Save a doctor without a name.	Save is blocked.	Performed as expected	Passed
	Doctor Contact Module	Doctor Settings	11/09/2026	CASE-062	Save a doctor with an invalid phone number.	Save is blocked.	Performed as expected	Passed
	Doctor Contact Module	Doctor Settings	11/09/2026	CASE-063	Save a valid doctor contact.	Doctor appears in Settings after save.	Performed as expected	Passed
	Data Management Module	Health Report Export	11/09/2026	CASE-064	Export health vitals without an older-adult profile.	Export is blocked and profile guidance appears.	Performed as expected	Passed
	Data Management Module	Health Report Export	11/09/2026	CASE-065	Export health vitals with no synchronized readings.	Empty-history guidance appears and no PDF is created.	Performed as expected	Passed
	Data Management Module	Health Report Export	11/09/2026	CASE-066	Export a synchronized health reading.	Patient-scoped recent history is sent to the PDF exporter.	Performed as expected	Passed
	Data Management Module	Health Report Export	11/09/2026	CASE-067	Return a database error while loading health readings.	Error is shown and no PDF is created.	Performed as expected	Passed
	Data Management Module	Medication Report Export	11/09/2026	CASE-068	Export an empty medication schedule.	Add-medication guidance appears and no PDF is created.	Performed as expected	Passed
	Data Management Module	Medication Report Export	11/09/2026	CASE-069	Export a saved medication schedule.	Caregiver- and patient-scoped schedules are sent to the PDF exporter.	Performed as expected	Passed
	Data Management Module	Medication Report Export	11/09/2026	CASE-070	Return a database error while loading medication schedules.	Error is shown and no PDF is created.	Performed as expected	Passed
	Data Management Module	Health Report Export	11/09/2026	CASE-071	Fail PDF generation after loading a health reading.	Caregiver sees the export error.	Performed as expected	Passed
	Doctor Contact Module	Doctor Settings	11/09/2026	CASE-072	Return a database error while saving a doctor contact.	Error appears and the edit form stays open.	Performed as expected	Passed
	Doctor Contact Module	Doctor Settings	11/09/2026	CASE-073	Edit a saved doctor contact.	Existing record is updated for the same caregiver and older adult.	Performed as expected	Passed
	Vitals	BPM display bands	11/09/2026	CASE-074	Compare 49, 50, 79, 80, 110 and 111 BPM.	Labels change at 50 and 80; warning remains strictly above 110.	Performed as expected	Passed
	Vitals	Other status labels	11/09/2026	CASE-075	Check saved oxygen, sleep, activity, HRV stress and skin temperature.	Each metric has a readable label with an appropriate state.	Performed as expected	Passed
	Medication	Time parsing	11/09/2026	CASE-076	Convert midnight, noon and database seconds into 12-hour time.	AM/PM values are correct and invalid times rejected.	Performed as expected	Passed
	Medication	Time zone boundaries	11/09/2026	CASE-077	Convert Manila medication times and reject DST gaps.	Saved wall clocks are not interpreted as UTC.	Performed as expected	Passed
	Care plan	Chronology	11/09/2026	CASE-078	Sort same-day dose times and appointment phases.	Events use actual date and time; completed status is distinct.	Performed as expected	Passed
	Analytics	Chart integrity	11/09/2026	CASE-079	Reduce 200 readings for display.	Original endpoints and extremes remain; no average is invented.	Performed as expected	Passed
	Analytics	Chart controls	11/09/2026	CASE-080	Switch line and thin bar display and inspect a reading.	Both chart modes and original timestamps are available.	Performed as expected	Passed
	Steps	Daily total	11/09/2026	CASE-081	Read cumulative step samples around Manila midnight.	One maximum per calendar day; samples are not summed.	Performed as expected	Passed

Proponent	Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Result	Actual Result	Result
	Reminders	Local ownership	11/09/2026	CASE-082	Deliver repeating reminders to account and patient scope.	Different delivery dates have distinct IDs; foreign scope is rejected.	Performed as expected	Passed
	Reminders	Popup queue	11/09/2026	CASE-083	Queue two events with one duplicate.	Each unique event appears once in sequence.	Performed as expected	Passed
	Medication	Prescription draft	11/09/2026	CASE-084	Normalize incomplete extracted prescription fields.	Unknown times, dose and prescriber stay blank.	Performed as expected	Passed
	Emergency	Contact priority	11/09/2026	CASE-085	Read contacts stored in an arbitrary order.	Caregiver, family member and doctor are always ordered 1 to 3.	Performed as expected	Passed
	Alerts	Collapsible rules	11/09/2026	CASE-086	Open and close automatic vital alerts.	Rules begin collapsed and are accessible when expanded.	Performed as expected	Passed
	Prescription	Elle entry	11/09/2026	CASE-087	Open optional prescription scan from Elle card.	Elle illustration and draft-review entry are visible.	Performed as expected	Passed
	Account history	Patient isolation	11/09/2026	CASE-088	Switch accounts and restore saved sleep.	Each patient sees only their own recorded sleep.	Performed as expected	Passed
	Administrator	Security	11/09/2026	CASE-089	Review roles and restricted actions.	Unauthorized changes are refused and audited.	Performed as expected	Passed
	Alerts	Response	11/09/2026	CASE-090	Acknowledge and resolve a saved warning.	Response note and status save once; failed saves can retry.	Performed as expected	Passed
	Medication images	Private storage	11/09/2026	CASE-091	Attempt an upload after the account changes.	Upload stops before private photo data is sent.	Performed as expected	Passed
	Notifications	Tap routing	11/09/2026	CASE-092	Tap a notification before session restoration.	Correct patient page opens after sign in.	Performed as expected	Passed
	Care plan	Save recovery	11/09/2026	CASE-093	Save medication or visit when local scheduling fails.	Committed record remains saved; draft survives failed database write.	Performed as expected	Passed
	Galaxy Watch6	Vital decoding	11/09/2026	CASE-094	Read multiple sensor metrics with timestamps.	Each valid metric reaches the correct card with its own time.	Performed as expected	Passed
	Health records	Saved history	11/09/2026	CASE-095	Load saved history, checkups and schedules.	Existing database records appear without fabricated values.	Performed as expected	Passed
	Live location	Map update	11/09/2026	CASE-096	Receive a fix before the map initializes.	Newest valid watch marker appears after map initialization.	Performed as expected	Passed
	Medication	Edit retry	11/09/2026	CASE-097	Correct a draft after a failed medication save.	Draft stays editable until save succeeds.	Performed as expected	Passed
	Health history	Samsung sync	11/09/2026	CASE-098	Resume interrupted sleep and oxygen uploads.	Checkpoint resumes the correct patient and source.	Performed as expected	Passed
	GPS tracking	Movement recovery	11/09/2026	CASE-099	Read saved positions when realtime events stop.	Only the selected patient and watch route updates.	Performed as expected	Passed
	SOS and warnings	Foreground alert	11/09/2026	CASE-100	Receive SOS and vital warnings.	Alert sounds once; duplicates and stale events are ignored.	Performed as expected	Passed

Each table ID matches one CASE-###.test.tsx file. Automated native and cloud boundaries are mocked where applicable. Device delivery, GPS accuracy, and push notification behavior also need a live demonstration.